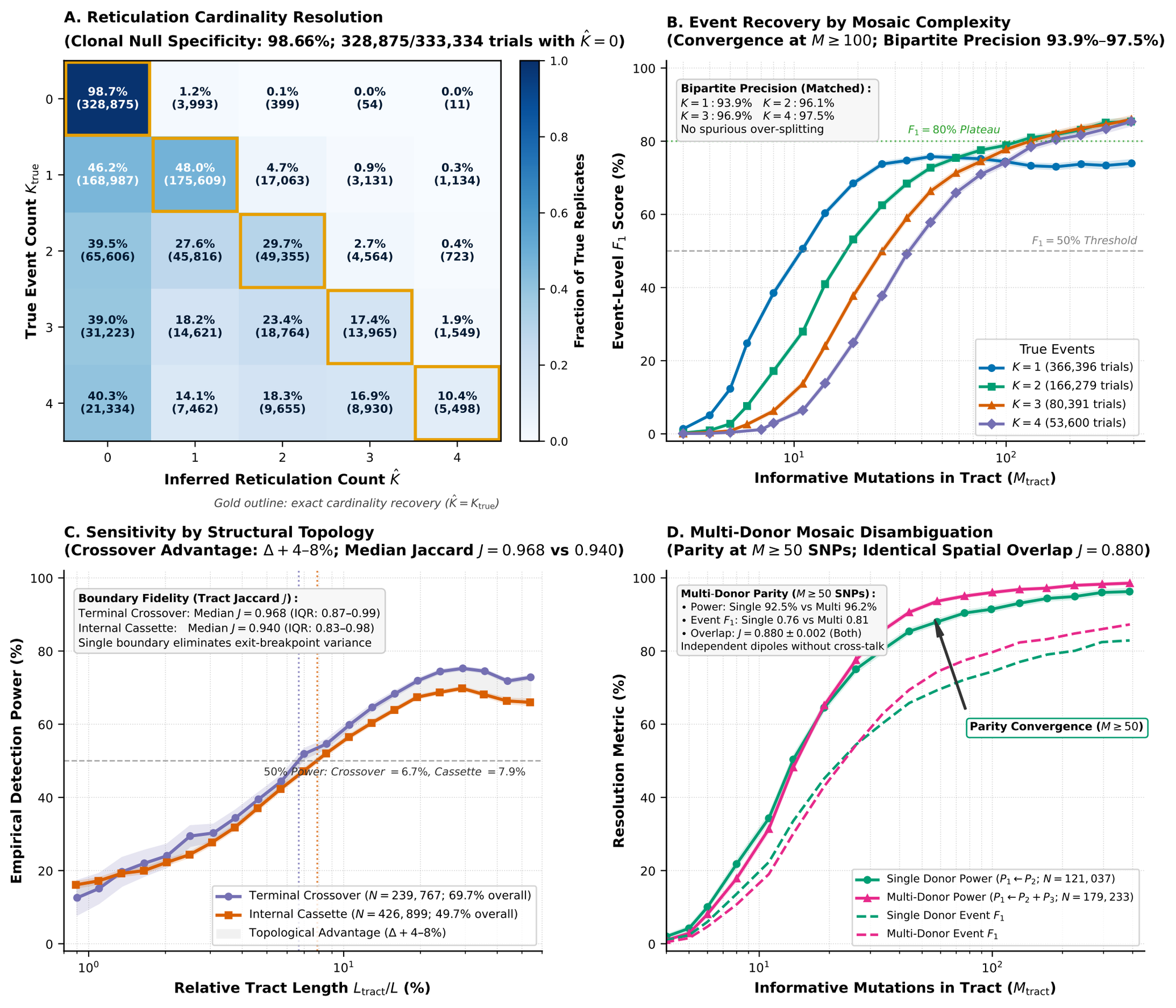

A. Reticulation Cardinality Resolution
(Clonal Null Specificity: 98.66%; 328,875/333,334 trials with $\hat{K} = 0$)
98.7% (328,875)
1.2% (3,993)
0.1% (399)
0.0% (54)
0.0% (11)
46.2% (168,987)
48.0% (175,609)
4.7% (17,063)
0.9% (3,131)
0.3% (1,134)
39.5% (65,606)
27.6% (45,816)
29.7% (49,355)
2.7% (4,564)
0.4% (723)
39.0% (31,223)
18.2% (14,621)
23.4% (18,764)
17.4% (13,965)
1.9% (1,549)
40.3% (21,334)
14.1% (7,462)
18.3% (9,655)
16.9% (8,930)
10.4% (5,498)
True Event Count K_{true}
Inferred Reticulation Count $\hat{K}$
Fraction of True Replicates
Gold outline: exact cardinality recovery ($\hat{K} = K_{true}$)
B. Event Recovery by Mosaic Complexity
(Convergence at $M \geq 100$; Bipartite Precision 93.9%–97.5%)
Bipartite Precision (Matched):
$K = 1$: 93.9% $K = 2$: 96.1%
$K = 3$: 96.9% $K = 4$: 97.5%
No spurious over-splitting
$F_1 = 80\%$ Plateau
$F_1 = 50\%$ Threshold
True Events
$K = 1$ (366,396 trials)
$K = 2$ (166,279 trials)
$K = 3$ (80,391 trials)
$K = 4$ (53,600 trials)
Event-Level F_1 Score (%)
Informative Mutations in Tract (M_{tract})
C. Sensitivity by Structural Topology
(Crossover Advantage: $\Delta + 4$–8%; Median Jaccard $J = 0.968$ vs 0.940)
Boundary Fidelity (Tract Jaccard J):
Terminal Crossover: Median $J = 0.968$ (IQR: 0.87–0.99)
Internal Cassette: Median $J = 0.940$ (IQR: 0.83–0.98)
Single boundary eliminates exit-breakpoint variance
50% Power: Crossover = 6.7%, Cassette = 7.9%
Terminal Crossover ($N = 239,767$; 69.7% overall)
Internal Cassette ($N = 426,899$; 49.7% overall)
Topological Advantage ($\Delta + 4$–8%)
Empirical Detection Power (%)
Relative Tract Length L_{tract}/L (%)
D. Multi-Donor Mosaic Disambiguation
(Parity at $M \geq 50$ SNPs; Identical Spatial Overlap $J = 0.880$)
Multi-Donor Parity ($M \geq 50$ SNPs):
• Power: Single 92.5% vs Multi 96.2%
• Event F_1: Single 0.76 vs Multi 0.81
• Overlap: $J = 0.880 \pm 0.002$ (Both)
Independent dipoles without cross-talk
Parity Convergence ($M \geq 50$)
Single Donor Power ($P_1 \leftarrow P_2$; $N = 121,037$)
Multi-Donor Power ($P_1 \leftarrow P_2 + P_3$; $N = 179,233$)
Single Donor Event F_1
Multi-Donor Event F_1
Resolution Metric (%)
Informative Mutations in Tract (M_{tract})